

GARCH and stochastic volatility models

Christian Francq

Chapter 5: Stochastic Volatility Models

Outline

- 1 General stochastic volatility model
 - Canonical stochastic volatility model
 - Estimation

- 2 Markov-switching models
 - Theory
 - Application

Pros and cons of GARCH models

- Advantages of GARCH :
 - Take into account some stylized facts : leptokurticity, volatility clustering, leverage effect.
 - For numerous GARCH-type models, the main probabilistic structure (stationarity, ergodicity, mixing, ...) is known (but the marginal distribution, for instance, is generally unknown).
 - The statistical inference of standard GARCH is relatively easy (but not for EGARCH, in particular).
- Drawbacks of GARCH :
 - The standard GARCH parameters appear to be very constrained (they drive simultaneously the decrease of the squared autocorrelations and the existence of moments).
 - The conditional second and fourth-order moments are time-varying but the conditional Kurtosis is constant.
 - Not able to handle changes of regimes (bear and bull market cycles).

General form of the stochastic volatility models

GARCH-type models $\varepsilon_t = \sqrt{h_t} \eta_t$

where h_t is a function of the past.

- Only the conditional variance is time-varying, but the shape of the conditional distribution is constant (Kurtosis, Skewness, ...)
- Marginal moments ($E\varepsilon_t^2 < \infty$ iff $\alpha + \beta < 1$) and persistence of the shocks ($\rho_{\varepsilon^2}(h) = (\alpha + \beta)\rho_{\varepsilon^2}(h - 1)$) are artificially linked.

Stochastic volatility models $\varepsilon_t = \sqrt{h_t} \eta_t$

where $h_t \notin \text{past of } \varepsilon_t \quad (\implies \quad h_t \neq E(\varepsilon_t^2 \mid \varepsilon_{t-1}))$

Canonical model

$$\begin{cases} \varepsilon_t &= \sqrt{h_t} \eta_t \\ \log h_t &= \omega + \beta \log h_{t-1} + \sigma v_t \end{cases}$$

where $|\beta| < 1$, (η_t) and (v_t) are two independent strong noises (with unit variance).

β is the parameter for the persistence of the shocks

ω is a scaling parameter for the volatility

σ is the volatility of the log-volatility

Stationarity

Strictly stationary solution iff $|\beta| < 1$

$$\varepsilon_t = \exp \left\{ \frac{\omega}{2(1-\beta)} + \frac{\sigma}{2} \sum_{i=0}^{\infty} \beta^i v_{t-i} \right\} \eta_t$$

The solution is a weak white noise (in particular $E\varepsilon_t^2 < \infty$) if

$$|\beta| < 1 \quad \text{and} \quad \prod_{i=0}^{\infty} \alpha_i < \infty, \quad \alpha_i = E\{\exp(\sigma\beta^i v_t)\}.$$

We then have

$$E\varepsilon_t^2 = e^{\frac{\omega}{1-\beta}} \prod_{i=0}^{\infty} \alpha_i.$$

Explicit expression when $v_t \sim \mathcal{N}(0, 1)$

We have $E(e^{\rho X}) = e^{\rho^2/2}$ if $X \sim \mathcal{N}(0, 1)$.

Therefore

$$\alpha_i = E\{\exp(\sigma\beta^i v_t)\} = \exp\left(\frac{\sigma^2}{2}\beta^{2i}\right),$$

$$\prod_{i=0}^{\infty} \alpha_i = \exp\left\{\frac{\sigma^2}{2(1-\beta^2)}\right\}$$

and

$$E\varepsilon_t^2 = \exp\left\{\frac{\omega}{1-\beta} + \frac{\sigma^2}{2(1-\beta^2)}\right\}$$

Higher-order moments

The moment of order r exists if

$$|\beta| < 1 \quad \text{and} \quad \prod_{i=0}^{\infty} \alpha_{i,r} < \infty, \quad \alpha_{i,r} = E\left\{\exp\left(\frac{r}{2}\sigma\beta^i v_t\right)\right\}.$$

We then have

$$E|\varepsilon_t|^r = E|\eta_t|^r e^{\frac{\omega r}{1-\beta}} \prod_{i=0}^{\infty} \alpha_{i,r}.$$

Autocovariances of (ε_t^2)

Letting, for $i \geq k$, $\alpha_{k,i} = E[\exp\{\sigma\beta^i(1 + \beta^{-k})v_t\}]$, we have

$$\text{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) = e^{\frac{2\omega}{1-\beta}} \left(\prod_{i=0}^{k-1} \alpha_i \prod_{i=k}^{\infty} \alpha_{k,i} - \prod_{i=0}^{\infty} \alpha_i^2 \right).$$

If $v_t \sim \mathcal{N}(0, 1)$,, for all lag $k > 0$

$$\text{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) = e^{\frac{2\omega}{1-\beta} + \frac{\sigma^2}{1-\beta^2}} \left(e^{\frac{\sigma^2}{1-\beta^2} \beta^k} - 1 \right).$$

The decrease of the autocovariances is not compatible with an ARMA model, but the rate is the same $\text{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) \sim cste \times \beta^k$ as $k \rightarrow \infty$.

ARMA(1,1) representation for $X_t = \log \varepsilon_t^2$

If $|\beta| < 1$ and $E |\ln \eta_t^2|^2 < \infty$, then

$$X_t - \mu_X = \beta(X_{t-1} - \mu_X) + u_t - \alpha u_{t-1},$$

where (u_t) is a weak white noise.

We have $X_t - \mu_X = Y_t + Z_t$ where the strong white noise $Z_t = \log \eta_t^2 - E \log \eta_t^2$ is independent of the AR(1) process $Y_t = \log h_t - E \log h_t$.

Since $\gamma_Y(h) = \beta \gamma_Y(h-1)$ for all $h > 0$, we have

$$\gamma_X(h) = \text{Cov}(Y_t + Z_t, Y_{t-h} + Z_{t-h}) = \gamma_Y(h) = \rho \gamma_X(h-1), \quad \forall h > 1,$$

which characterizes a (weak) ARMA(1, 1).

Identification

$$\begin{cases} \varepsilon_t &= \sqrt{h_t} \eta_t \\ \log h_t &= \omega + \beta \log h_{t-1} + \sigma v_t, \end{cases} \quad |\beta| < 1.$$

Without loss of generality, assume that $\sigma \geq 0$ and $\sigma \neq 0$ (otherwise the process $\log h_t$ is deterministic, and ω and β are not identifiable). Thus

$$\sigma > 0.$$

The model is unchanged when η_t is replaced by $c\eta_t$ ($c > 0$) and ω by $\omega - 2(1 - \beta) \log c$. One can thus assume

$$V(\eta_t) = E(\eta_t^2) = 1 \quad (\text{if } E\eta_t^2 < \infty)$$

or (alternatively) that for a given number μ

$$E(\log \eta_t^2) = \mu \quad (\text{if } E|\log \eta_t^2| < \infty).$$

Estimation methods

- **Moments** Taylor (1986) ;
- **Generalized method of moments (GMM)** Melino and Turnbull (1990), Andersen and Sørensen (1996) ;
- **Quasi-maximum likelihood (QML)** Nelson (1988), Harvey, Ruiz and Shephard (1994) and Ruiz (1994) ;
- **MCMC** Jacquier, Polson and Rossi (1994), Chib, Nardari and Shephard (2002) ;
- **Linear representations (ARMA)** Francq and Zakoïan (2006).

State-space representation : observation equation + state equation

Measurement equation + transition equation

$$\begin{cases} y_t &= U_t \alpha_t + d_t + u_t \\ \alpha_t &= T_t \alpha_{t-1} + c_t + R_t v_t, \quad t \geq 1 \end{cases}$$

- U_t, d_t, T_t, c_t and R_t not random,
- y_t observable vector,
- α_t unobservable *state vector*.
- $Eu_t = 0, Ev_t = 0, (u_t) \perp, (v_t) \perp, (u_t) \perp (v_t),$

Examples

Random walk plus noise

$$\begin{aligned}y_t &= \mu_t + \epsilon_t \\ \mu_t &= \mu_{t-1} + \eta_t\end{aligned}$$

AR(2)

$$\begin{aligned}y_t &= [1, 0]\alpha_t \\ \alpha_t &= \begin{bmatrix} y_t \\ y_{t-1} \end{bmatrix} = \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix} \alpha_{t-1} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \epsilon_t\end{aligned}$$

MA(1)

$$\begin{aligned}y_t &= [1, 0]\alpha_t \\ \alpha_t &= \begin{bmatrix} \eta_t + \theta\eta_{t-1} \\ \theta\eta_t \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \alpha_{t-1} + \begin{bmatrix} 1 \\ \theta \end{bmatrix} \eta_t\end{aligned}$$

State-space representation of the VS model

$$\begin{cases} \log \varepsilon_t^2 &= \log h_t + \mu_Z + u_t \\ \log h_t &= \beta \log h_{t-1} + \omega + \sigma v_t \end{cases}$$

with $Z_t = \log \eta_t^2$, $\mu_Z = E(Z_t)$, $u_t = Z_t - \mu_Z$ and $\sigma_Z^2 = \text{Var}(\log \eta_t^2)$, when existing.

Principle of the Kalman filter

$$\begin{cases} y_t &= U_t \alpha_t + d_t + u_t \\ \alpha_t &= T_t \alpha_{t-1} + c_t + R_t v_t \end{cases}$$

Objectives :

prediction $\alpha_{t|t-1} = E(\alpha_t | y_1, \dots, y_{t-1}),$
 $P_{t|t-1} = \text{Var}(\alpha_t | y_1, \dots, y_{t-1}),$

filtering $\alpha_{t|t} = E(\alpha_t | y_1, \dots, y_t),$
 $P_{t|t} = \text{Var}(\alpha_t | y_1, \dots, y_t),$

smoothing $E(\alpha_t | y_1, \dots, y_n).$

If $u_t \rightsquigarrow \mathcal{N}(0, H_t)$, $v_t \rightsquigarrow \mathcal{N}(0, Q_t)$, $\alpha_0 \rightsquigarrow \mathcal{N}(a_0, P_0)$, $\alpha_0 \perp (u_t), (v_t)$
then the law of α_t given $y_1, \dots, y_{t-1}$ is gaussian, with mean

the optimal predictor $\alpha_{t|t-1}$.

Applying the Kalman filter

$$\begin{cases} y_t &= U_t \alpha_t + d_t + u_t \\ \alpha_t &= T_t \alpha_{t-1} + c_t + R_t v_t \end{cases}$$

First step : prediction equations

$$\alpha_{t|t-1} = T_t \alpha_{t-1|t-1} + c_t, \quad P_{t|t-1} = T_t P_{t-1|t-1} T_t' + R_t Q_t R_t'.$$

taking the conditional mean and variance with respect to $y_1, \dots, y_{t-1}$ in the transition equation.

Valeurs initiales : $\alpha_{1|0} = T_1 a_0 + c_1$, $P_{1|0} = T_1 P_0 T_1' + R_1 Q_1 R_1'$.

Applying the Kalman filter

$$\begin{cases} y_t &= U_t \alpha_t + d_t + u_t \\ \alpha_t &= T_t \alpha_{t-1} + c_t + R_t v_t \end{cases}$$

Second step : updating

$$\alpha_{t|t} = \alpha_{t|t-1} + P_{t|t-1} U_t' F_t^{-1} (y_t - U_t \alpha_{t|t-1} - d_t)$$

$$P_{t|t} = P_{t|t-1} - P_{t|t-1} U_t' F_t^{-1} U_t P_{t|t-1}$$

setting $F_t := \text{Var}(y_t | y_1, \dots, y_{t-1}) = U_t P_{t|t-1} U_t' + H_t$.

We use the fact that the distribution of (y_t, α_t) given $y_{t-1}, \dots, y_1$ is gaussian, to find the distribution of α_t given $y_t, \dots, y_1$

► gaussian regression

Kalman filter : application to VS models

$$\alpha_{t|t-1} = \beta \alpha_{t-1|t-1} + \omega, \quad P_{t|t-1} = \beta^2 P_{t-1|t-1} + \sigma^2$$

$$\alpha_{t|t} = \alpha_{t|t-1} + P_{t|t-1} F_t^{-1} (\log \varepsilon_t^2 - \alpha_{t|t-1} - \mu_Z)$$

$$\alpha_{1|0} = \omega / (1 - \beta), \quad P_{1|0} = \sigma^2 / (1 - \beta^2)$$

and $F_t = P_{t|t-1} + \sigma_Z^2$, $P_{t|t} = P_{t|t-1} - P_{t|t-1}^2 F_t^{-1}$

(pseudo) log-likelihood :

$$\log \ell(\varepsilon_1, \dots, \varepsilon_n; \theta) \stackrel{c}{=} -\frac{1}{2} \sum_{t=1}^n \log F_t + \frac{(\log(\varepsilon_t^2) - \alpha_{t|t-1} - \mu_Z)^2}{F_t}$$

Generalized Method of Moments (GMM)

This method has been introduced by Hansen (1982).

Let (ε_t) a stationary process with a distribution depending on $\theta \in \mathbb{R}^k$.

We assume there exists a $m \times 1$ vector $g(\varepsilon_t, \theta)$ such that :

$$g_0(\theta) := E_{\theta_0}\{g(\varepsilon_t, \theta)\} = 0 \quad \text{if and only if} \quad \theta = \theta_0.$$

It is said that $E_{\theta_0}\{g(\varepsilon_t, \theta)\}$ is the **vector of the generalized moments**. In general $m > k$.

GMM Estimator

The vector $g_0(\theta)$ can be empirically estimated by

$$g_n(\theta) = \frac{1}{n} \sum_{t=1}^n g(\varepsilon_t, \theta).$$

Any norm of this vector should be asymptotically small at θ_0 . A **GMM estimator** of θ_0 is a solution of

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} g_n(\theta)' \Omega_n g_n(\theta)$$

where Ω_n is a $m \times m$ symmetric definite-positive matrix, which can depends of the observations.

Asymptotic properties of the GMM estimator

Assume that (i) $\Omega_n \rightarrow \Omega$ a.s., where Ω is a non random definite positive matrix, (ii) $E_{\theta_0} \sup_{\theta \in \Theta} \|g(\epsilon_t, \theta)\| < \infty$ and (iii) $\theta \rightarrow g(x, \theta)$ is continuous for all x .

Consistency

Under the previous assumptions, $\hat{\theta}_n \rightarrow \theta_0$ a.s. when $n \rightarrow \infty$.

Under additional regularity conditions

Asymptotic normality : $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}\{0, \Sigma(\Omega)\}$,

where

$$\Sigma(\Omega) = \{G\Omega G'\}^{-1} G\Omega V\Omega G' \{G\Omega G'\}^{-1},$$

$$G = E_{\theta_0} \frac{\partial g'(\epsilon_t, \theta_0)}{\partial \theta} \quad \text{and} \quad V = \sum_{h=-\infty}^{\infty} \text{Cov}_{\theta_0} \{g(\epsilon_t, \theta_0), g(\epsilon_{t-h}, \theta_0)\}.$$

(long term variance of $g(\epsilon_t, \theta_0)$)

Optimal choice of Ω (for a given vector of moments)

GMM optimal : obtained for $\Omega^* = V^{-1}$

$$\{GV^{-1}G'\}^{-1} = \Sigma(V^{-1}) \ll \Sigma(\Omega), \quad \forall \Omega \text{ symmetric definite-positive.}$$

2-step implementation :

- ① Define an initial GMM $\hat{\theta}_n^{(0)}$ by $\Omega_n = I_m$ or $\Omega_n = \Omega^{(0)}$, with $\Omega^{(0)}$ a fixed matrix of weights.
- ② From $g(\epsilon_t, \hat{\theta}_n^{(0)})$ $t = 1, \dots, n$, determine an estimate $\hat{V}$ of V .
An asymptotically optimal GMM is obtained with $\Omega_n = \hat{V}^{-1}$

Application to the SV model

The distributions of η_t and v_t are assumed to be known. One can take the moments conditions

$$E_{\theta_0}[g(\varepsilon_t, \dots, \varepsilon_{t-K}, \theta_0)] = 0,$$

with, for instance,

$$g(\varepsilon_t, \dots, \varepsilon_{t-K}, \theta) = \begin{pmatrix} \varepsilon_t^2 - E_{\theta} \varepsilon_t^2 \\ |\varepsilon_t| - E_{\theta} |\varepsilon_t| \\ \vdots \\ |\varepsilon_t|^r - E_{\theta} |\varepsilon_t|^r \\ |\varepsilon_t \varepsilon_{t-1}| - E_{\theta} |\varepsilon_t \varepsilon_{t-1}| \\ \vdots \end{pmatrix}$$

Motivation

When GARCH models are fitted on sub-periods of a long series, **important differences** can be observed on the estimated parameters.

These differences suggest **changes in dynamics**. There is empirical evidence for the existence of different regimes, in particular bull and bear market cycles, periods of upward and downward market trends.

A relatively easy way to take into account these regime changes is to assume that there are driven by a **hidden** (*i.e.* not observed) **Markov chain**.

- **Hidden Markov Model (HMM)** Baum et Petrie (1966) in the discrete case, Leroux (1992) in the general case ;
- **Markov-switching models** Hamilton (1989)

Hidden Markov chain

► Markov chain

$(X_t)_{t \geq 0}$ **hidden Markov chain** if

- ① conditionally to the non observed Markov chain (Δ_t) , the observations $X_0, X_1, \dots$ are independent ;
- ② the distribution of X_s given (Δ_t) depends only on Δ_s .

Example

$$\begin{cases} \varepsilon_t &= \sqrt{h_t} \eta_t, \\ h_t &= \omega(\Delta_t), \end{cases}$$

where (Δ_t) is a Markov chain on $\mathcal{E} = \{1, \dots, d\}$, irreducible and aperiodic, independent of the strong white noise (η_t)

$$\omega(1) < \dots < \omega(d).$$

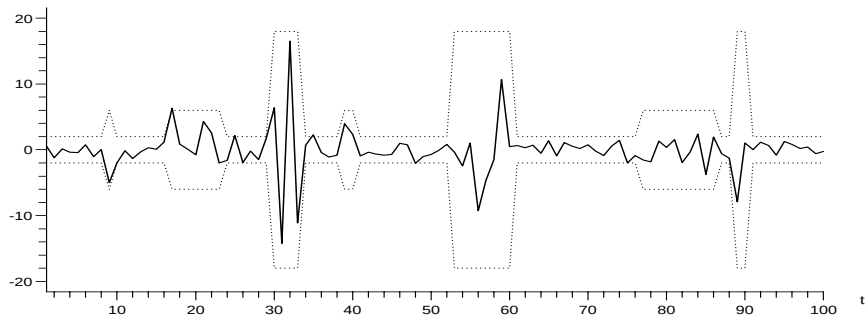


Figure – Simulation of length 100 and $\pm 2\omega(\Delta_t)$.

Properties of the HMM $\varepsilon_t = \sqrt{\omega(\Delta_t)}\eta_t$

- Volatility clustering,
- Leptokurticity,
- Non constant conditional Kurtosis

$$\frac{E(\varepsilon_t^4 | \varepsilon_{t-1}, \varepsilon_{t-2}, \dots)}{[E(\varepsilon_t^2 | \varepsilon_{t-1}, \varepsilon_{t-2}, \dots)]^2} = \frac{E(h_t^2 | \varepsilon_{t-1}, \varepsilon_{t-2}, \dots)}{[E(h_t | \varepsilon_{t-1}, \varepsilon_{t-2}, \dots)]^2} \kappa_\eta$$

- Possibility of high-order moments for ε_t ,
- Possibility of strong persistence

$$\text{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) = \lambda^k \{\omega(1) - \omega(2)\}^2 \pi(1)\pi(2), \quad k > 0$$

for $d = 2$ and $\lambda := p(1, 1) + p(2, 2) - 1$.

Markov-switching ARCH(1)

$$\begin{cases} \varepsilon_t &= \sqrt{h_t} \eta_t, \\ h_t &= \omega(\Delta_t) + \alpha(\Delta_t) \varepsilon_{t-1}^2 \end{cases}$$

with the same assumption on (Δ_t) and (η_t) . This model is no more a HMM.

Stationarity conditions ?

$$\begin{aligned} h_t &= \omega(\Delta_t) + \alpha(\Delta_t) \eta_{t-1}^2 h_{t-1} \\ &= \omega(\Delta_t) + a(\Delta_t, \eta_{t-1}) h_{t-1} \\ &= \omega(\Delta_t) + \sum_{k=1}^{\infty} a(\Delta_t, \eta_{t-1}) \dots a(\Delta_{t-k+1}, \eta_{t-k}) \omega(\Delta_{t-k}) \end{aligned}$$

if the series converges a.s.

Markov-switching ARCH(1) : strict stationarity

Strict stationarity condition

$$E \log \alpha(\Delta_t) = \sum_{i=1}^d \pi(i) \log \alpha(i) < E(-\log \eta_t^2).$$

Remarks :

- Certain regimes can be explosive (if they are not too often present)
- If $\alpha(i) = 0$ in one regime, the process is always strictly stationary
- The condition does not depend on ω

Markov-switching ARCH(1) : second-order stationarity

$$E\{\alpha(\Delta_t) \dots \alpha(\Delta_{t-k+1})\omega(\Delta_{t-k})\} = (1, \dots, 1) \mathbb{P}^{k+1}(\alpha) \boldsymbol{\pi}(\omega)$$

where

$$\mathbb{P}(\alpha) = \begin{pmatrix} \alpha(1)p(1,1) & \alpha(1)p(2,1) & \dots & \alpha(1)p(d,1) \\ \vdots & & & \\ \alpha(d)p(1,d) & \alpha(d)p(2,d) & \dots & \alpha(d)p(d,d) \end{pmatrix}, \quad \boldsymbol{\pi}(\omega) = \begin{pmatrix} \pi(1)\omega(1) \\ \vdots \\ \pi(d)\omega(d) \end{pmatrix}$$

Second-order stationarity condition :

$$\rho(\mathbb{P}(\alpha)) < 1.$$

Likelihood of the HMM $\varepsilon_t = \sqrt{\omega(\Delta_t)}\eta_t$, $\eta_t \sim \mathcal{N}(0, 1)$

Parameters $p(i, j) := P(\Delta_t = j \mid \Delta_{t-1} = i)$

d regimes $\omega(i)$ + $d(d-1)$ transition probabilities $p(i, j)$

Likelihood

$$L(\varepsilon_1, \dots, \varepsilon_n) = \sum_{(e_1, \dots, e_n) \in \mathcal{E}^n} L^{(e_1, \dots, e_n)}(\varepsilon_1, \dots, \varepsilon_n) P(e_1, \dots, e_n),$$

$$\begin{aligned} P(e_1, \dots, e_n) &= \pi(e_1) p(e_1, e_2) \dots p(e_{n-1}, e_n) \\ L^{(e_1, \dots, e_n)}(\varepsilon_1, \dots, \varepsilon_n) &= \prod_{t=1}^n \phi_{e_t}(\varepsilon_t), \quad \phi_i(\cdot) \sim \mathcal{N}\{0, \omega(i)\} \end{aligned}$$

but $2^{300} >$ total number of atoms in the universe !

Computation of the likelihood

- Matrix form
- Forward-Backward algorithm
- Hamilton's filter

Matrix form of the likelihood

$$L(\varepsilon_1, \dots, \varepsilon_n) = \mathbf{1}' M(\varepsilon_n) M(\varepsilon_{n-1}) \cdots M(\varepsilon_2) F_1,$$

[▶ Proof](#)

$$M(x) = \begin{pmatrix} p(1,1)\phi_1(x) & \cdots & p(d,1)\phi_1(x) \\ \vdots & & \\ p(1,d)\phi_d(x) & \cdots & p(d,d)\phi_d(x) \end{pmatrix}$$
$$F_1 = \begin{pmatrix} \pi(1)\phi_1(\varepsilon_1) \\ \vdots \\ \pi(d)\phi_d(\varepsilon_1) \end{pmatrix}.$$

This form of the likelihood is computable (of order nd^2 multiplications).

Markov-switching GARCH

Markov-switching GARCH(p, q) process

$$\begin{cases} \varepsilon_t &= \sqrt{h_t} \eta_t \\ h_t &= \omega(\Delta_t) + \sum_{i=1}^q \alpha_i(\Delta_t) \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j(\Delta_t) h_{t-j} \end{cases}$$

The estimation is complicated when $p \neq 0$.

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Standard GARCH(1, 1)

$$h_t = 0.13 + 0.184\varepsilon_{t-1}^2 + 0.739h_{t-1}$$

Strong persistence $0.184 + 0.739 = 0.923 \simeq 1$

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2 constant regimes (HMM)

$$h_t = \begin{cases} 0.81 & \text{if } \Delta_t = 1 \\ 4.46 & \text{if } \Delta_t = 2 \end{cases}, \quad \mathbb{P} = \begin{pmatrix} 0.977 & 0.023 \\ 0.102 & 0.898 \end{pmatrix}$$

Very different and persistent regimes

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MS-ARCH(1) with 2 regimes

$$h_t = \begin{cases} 0.76 + 0.05\varepsilon_{t-1}^2 & \text{if } \Delta_t = 1 \\ 3.71 + 0.10\varepsilon_{t-1}^2 & \text{if } \Delta_t = 2 \end{cases}, \quad \mathbb{P} = \begin{pmatrix} 0.979 & 0.021 \\ 0.084 & 0.916 \end{pmatrix}$$

Small ARCH effects in the two regimes

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MS-ARCH(2) with 2 regimes

$$h_t = \begin{cases} 0.76 + 0.05\varepsilon_{t-1}^2 + 0.00\varepsilon_{t-2}^2 & \text{if } \Delta_t = 1 \\ 3.70 + 0.10\varepsilon_{t-1}^2 + 0.00\varepsilon_{t-2}^2 & \text{if } \Delta_t = 2 \end{cases},$$
$$\mathbb{P} = \begin{pmatrix} 0.979 & 0.021 \\ 0.084 & 0.916 \end{pmatrix}$$

Identical to the MS-ARCH(1) with 2 regimes

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HMM with 3 regimes

$$h_t = \begin{cases} 0.42 & \text{if } \Delta_t = 1 \\ 1.27 & \text{if } \Delta_t = 2 \\ 6.77 & \text{if } \Delta_t = 3 \end{cases}, \mathbb{P} = \begin{pmatrix} 0.972 & 0.015 & 0.013 \\ 0.008 & 0.982 & 0.010 \\ 0.038 & 0.080 & 0.882 \end{pmatrix}$$

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MS-ARCH(1) with 3 regimes

$$h_t = \begin{cases} 0.41 + 0.01\varepsilon_{t-1}^2 & \text{si } \Delta_t = 1 \\ 1.25 + 0.00\varepsilon_{t-1}^2 & \text{si } \Delta_t = 2 \\ 5.75 + 0.09\varepsilon_{t-1}^2 & \text{si } \Delta_t = 3 \end{cases},$$
$$\mathbb{P} = \begin{pmatrix} 0.973 & 0.014 & 0.013 \\ 0.007 & 0.983 & 0.010 \\ 0.037 & 0.071 & 0.892 \end{pmatrix}$$

Small ARCH effects within each regime

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MS-ARCH(2) with 3 regimes

$$h_t = \begin{cases} 0.41 + 0.01\varepsilon_{t-1}^2 + 0.00\varepsilon_{t-2}^2 & \text{if } \Delta_t = 1 \\ 1.25 + 0.00\varepsilon_{t-1}^2 + 0.00\varepsilon_{t-2}^2 & \text{if } \Delta_t = 2 \\ 5.75 + 0.09\varepsilon_{t-1}^2 + 0.00\varepsilon_{t-2}^2 & \text{if } \Delta_t = 3 \end{cases},$$

$$\mathbb{P} = \begin{pmatrix} 0.973 & 0.014 & 0.013 \\ 0.007 & 0.983 & 0.010 \\ 0.037 & 0.071 & 0.892 \end{pmatrix}$$

Exactly the same as the MS-ARCH(1) with 3 regimes

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MS-ARCH(1) with 4 regimes

$$h_t = \begin{cases} 0.40 + 0.01\varepsilon_{t-1}^2 & \text{if } \Delta_t = 1 \\ 1.17 + 0.00\varepsilon_{t-1}^2 & \text{if } \Delta_t = 2 \\ 2.90 + 0.23\varepsilon_{t-1}^2 & \text{if } \Delta_t = 3 \\ 12.11 + 0.06\varepsilon_{t-1}^2 & \text{if } \Delta_t = 4 \end{cases},$$
$$\mathbb{P} = \begin{pmatrix} 0.972 & 0.016 & 0.009 & 0.003 \\ 0.006 & 0.983 & 0.001 & 0.010 \\ 0.000 & 0.028 & 0.972 & 0.000 \\ 0.178 & 0.139 & 0.000 & 0.682 \end{pmatrix}$$

Quite different volatilities, Regime 3 with some ARCH effect

Regime 4 less persistent

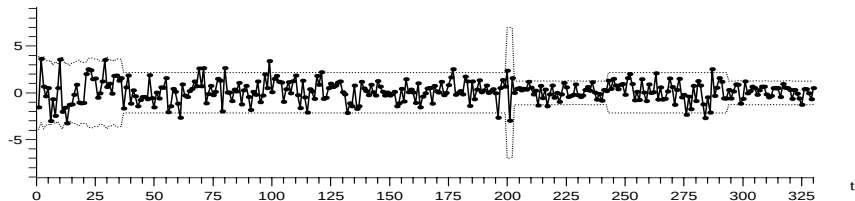
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HMM with 4 regimes

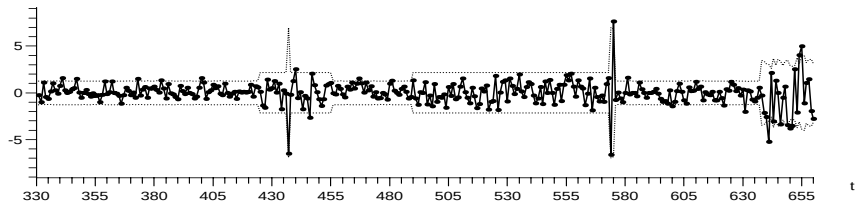
$$h_t = \begin{cases} 0.44 & \text{si } \Delta_t = 1 \\ 1.13 & \text{si } \Delta_t = 2 \\ 2.73 & \text{si } \Delta_t = 3 \\ 10.88 & \text{si } \Delta_t = 4 \\ 10.88 & \text{si } \Delta_t = 5 \end{cases}$$

One of the regimes is useless

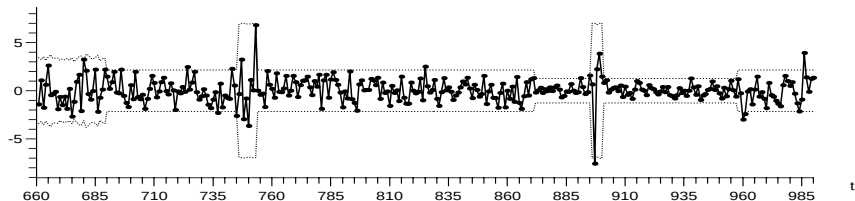
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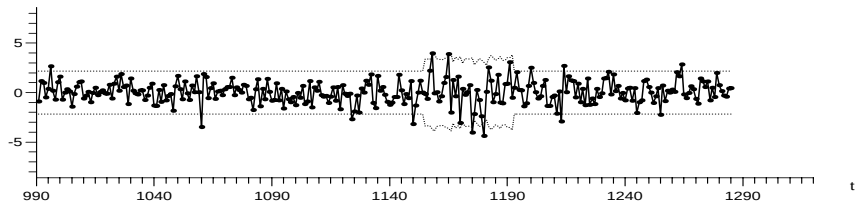
CAC 40 from 2 June 1988 to 31 December 1993



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End of Chapter 4

Conditional distribution for a gaussian vector

If the vector $(x, y)'$ is gaussian with

$$\mu_x = E(x), \mu_y = E(y),$$

$$\Sigma_{xx} = \text{Var}(x), \Sigma_{yy} = \text{Var}(y), \Sigma_{xy} = \Sigma'_{yx} = \text{Cov}(x, y),$$

the distribution of x conditional to y is

$$\mathcal{N}(\mu_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y - \mu_y), \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx})$$

◀ Return

Finite-state Markov chain

Markov chain : process such that the future depends only on the present and not of the past.

Definition

An homogeneous Markov chain $(\Delta_t)_{t \geq 0}$ valued in $\mathcal{E} = \{1, \dots, d\}$ is defined by :

- ① an **initial distribution** $\pi_0(i) = P(\Delta_0 = i)$ for $i = 1, \dots, d$ (with $\sum_{i=1}^d \pi_0(i) = 1$) ;
- ② a matrix of **transition probabilities** $\mathbb{P} = (p(i, j))_{1 \leq i, j \leq d}$ such that

$$\begin{aligned} p(i, j) &= P(\Delta_t = j | \Delta_{t-1} = i) \\ &= P(\Delta_t = j | \Delta_{t-1} = i, \Delta_{t-2} = e_{t-2}, \dots, \Delta_0 = e_0) \end{aligned}$$

(with $\sum_{j=1}^d p(i, j) = 1$ for all $i \in \mathcal{E}$).

Multi-step transitions

Let $\mathbb{P}^{(k)} = (p^{(k)}(i, j))_{1 \leq i, j \leq d}$ be the matrix of the k -step transitions :

$$p^{(k)}(i, j) = P(\Delta_t = j | \Delta_{t-k} = i).$$

We have

$$\mathbb{P}^{(k)} = \mathbb{P}^k$$

More generally :

$$\text{Chapman-Kolmogorov equations : } \mathbb{P}^{(k+k')} = \mathbb{P}^{(k)} \mathbb{P}^{(k')}.$$

Invariant distribution

Note that

$$\pi_0 = \begin{pmatrix} \pi_0(1) \\ \vdots \\ \pi_0(d) \end{pmatrix} \quad \text{and} \quad \pi_n = \begin{pmatrix} P(\Delta_n = 1) \\ \vdots \\ P(\Delta_n = d) \end{pmatrix}.$$

It can be easily seen that

$$\pi_n = \mathbb{P}' \pi_{n-1}, \quad \pi_n = \mathbb{P}'^n \pi_0, \quad n \geq 0.$$

A probability π on $\mathcal{E}$ is said to be an **invariant law** if

$$\pi = \mathbb{P}' \pi, \quad \pi' \mathbf{1} = 1 \quad (\text{with } \mathbf{1}' = (1, \dots, 1)).$$

If the limit law $\pi_\infty := \lim_{n \rightarrow \infty} \pi_n$ exists then this is an invariant distribution.

Irreducibility, aperiodicity

Two states i and j **communicate** if one can move from i to j in a finite number of states (i.e. $p^{(m)}(i, j) > 0$) if one can also move from j to i in a finite number of states. If all the states communicate, the chain is said to be **irreducible**.

A state i is said to be **aperiodic** if 1 is the greatest common divisor of the integers m such that $p^{(m)}(i, i) > 0$. The chain is called aperiodic if all the states are aperiodic.

All finite state-space Markov chain which is **irreducible and aperiodic** admits a **unique invariante distribution**. If the initial distribution is the invariant law, then the chain is stationary and ergodic.

[← Return](#)

Proof of the matrix form of the likelihood

Let

$$F_k(i) = g_k(\epsilon_1, \dots, \epsilon_k | \Delta_k = i) \pi(i)$$

where $g_k(\cdot | \Delta_k = i)$ is the law of $(\epsilon_1, \dots, \epsilon_k)$ given $\{\Delta_k = i\}$. One can easily check that

$$F_1(i) = \pi(i) \phi_i(\epsilon_1),$$

$$F_k(i) = \phi_i(\epsilon_k) \sum_{j=1}^d F_{k-1}(j) p(j, i).$$

In matrix form we get

$$F_k := (F_k(1), \dots, F_k(d))' = M(\epsilon_k) F_{k-1}.$$

Moreover

$$L_\theta(\epsilon_1, \dots, \epsilon_n) = \sum_{i=1}^d F_n(i).$$